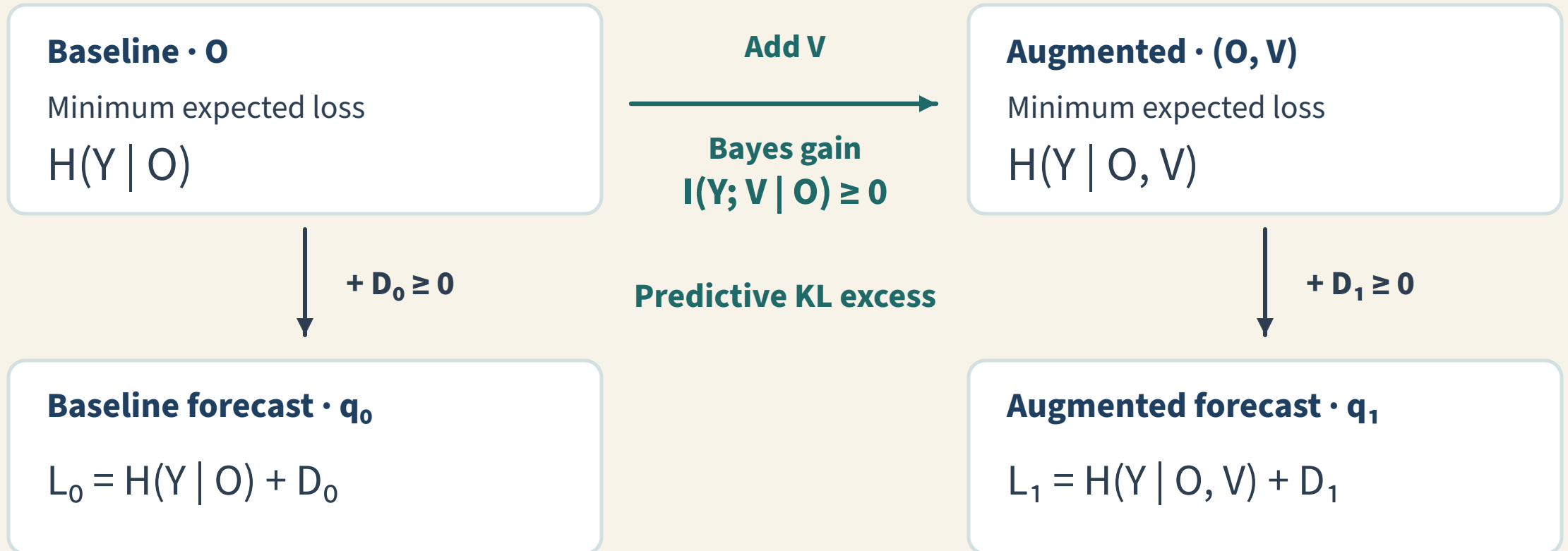


Information gain and expected model loss

One fixed finite law P · admissible forecasts q_0, q_1 · natural logs · nats

Bayes-optimal loss from the available observations



Expected loss reduction for these forecasts

$$L_0 - L_1 = I(Y; V \mid O) + D_0 - D_1$$

This can be negative even when the added observation has positive CMI.